

Data $S = \{ (x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(N)}, y^{(N)}) \}$

sampled from an unknown function

$$x^{(i)} \in \mathbb{R}^d$$

$$y^{(i)} \in \mathbb{R}$$

$$y = \underbrace{f^*(x)}_{\text{unknown true data generating function}} + \underbrace{\varepsilon}_{\text{unknown additive zero mean noise}}$$

Regression Model (Linear)

$$y = f(x, w) + \varepsilon$$

$$= w_0 x_0 + w_1 x_1 + \dots + w_d x_d + \varepsilon$$

$$\Rightarrow \hat{y} = w^T x$$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}, x_0 = 1$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}$$

Under this linear regression model,
the Maximum ^{Log}Likelihood Estimate (MLE) is:

$$w_{MLE} := \arg \max_w - \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2$$

which is the same as minimizing the Mean Squared Error

$$\hat{w} = w_{MLE} = \arg \min_w \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2$$

$$= \arg \min_w \frac{1}{N} \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2$$

$$= \arg \min_w \underbrace{\frac{1}{2N} \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2}_{L(w; S) \rightarrow \text{loss as a function of } w \text{ given dataset } S}$$

Computing gradients

$$\nabla_w L = \frac{1}{2N} \sum_{i=1}^N \frac{\partial}{\partial w} (y^{(i)} - w^T x^{(i)})^2$$

$$= \frac{1}{2N} \sum_{i=1}^N 2(y^{(i)} - w^T x^{(i)}) \cdot \frac{\partial}{\partial w} (y^{(i)} - w^T x^{(i)})$$

$$= - \frac{1}{N} \sum_{i=1}^N (y^{(i)} - w^T x^{(i)}) \cdot x^{(i)}$$

$$\nabla_w L(w) = -\frac{1}{N} \sum_{i=1}^N \underbrace{(y^{(i)} - w^T x^{(i)})}_{\text{scalar quantity measuring prediction error}} x^{(i)}$$

input data vector

Note: The gradient vector is a linear combination of the input data vectors. This aspect is crucial when we discuss kernel methods.

GRADIENT DESCENT

We start with some arbitrary value of $w = w(t)$, $t=0$ and use the gradient of the loss $\nabla_w L(w)$ to update w over the next several iterations. We weight the gradient with a learning rate η (also called as the step size) to obtain the update step:

$$w(t+1) = w(t) \ominus \eta \nabla_w L(w)$$

-ve sign because we are trying to minimize the loss fn. $L(w)$

$$w(t+1) = w(t) - \eta \left(-\frac{1}{N} \sum_{i=1}^N (y^{(i)} - w(t)^T x^{(i)}) x^{(i)} \right)$$

$$w(t+1) = w(t) + \eta \frac{1}{N} \sum_{i=1}^N (y^{(i)} - w(t)^T x^{(i)}) x^{(i)}$$

This update step is identical to the gradient ascent steps you would derive for maximizing the log likelihood fn.

There is equivalence between using the linear regression model (linear map + zero mean Gaussian noise) & maximizing the likelihood and minimizing the Mean Squared Error (MSE)

We can similarly derive the MLE estimate for logistic regression under the Bernoulli Distribution model.

An Alternative to gradient descent for linear regression

$$L(w) = \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2$$

Define

$$\vec{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{bmatrix}_{N \times 1}$$

$$X = \begin{bmatrix} x^{(1)} & x^{(2)} & \dots & x^{(N)} \end{bmatrix}_{d \times N}$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}_{d \times 1}$$

Then we can write

$$L(w) = \|\vec{y} - X^T w\|_2^2$$

$\|\cdot\|_2$ - L_2 norm

$$= (\vec{y} - X^T w)^T (\vec{y} - X^T w)$$

$$= \vec{y}^T \vec{y} - \underbrace{w^T X \vec{y} - \vec{y}^T X^T w}_{\text{scalars } \therefore \text{ same qty}} + w^T X X^T w$$

$$= \vec{y}^T \vec{y} - 2w^T X \vec{y} + w^T X X^T w$$

Differentiating w.r.t w , we get.

$$\nabla_w L(w) = \frac{\partial}{\partial w} (w^T X X^T w - 2w^T X \vec{y})$$

$$= 2X X^T w - 2X \vec{y}$$

Setting $\nabla_w L(w) = 0$, we get

$$X X^T w = X \vec{y}$$

$$\hat{w} = (X X^T)^{-1} X \vec{y}$$

Assuming
 $X X^T$ is invertible

When inverting $X X^T$ is too expensive, use iterative gradient updates.

Linear Regression with Regularization

$$L_{\text{ridge}}(w) = \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2 + \lambda \|w\|_2^2$$

$$= \vec{y}^T \vec{y} - 2 \vec{w}^T X \vec{y} + \vec{w}^T X X^T \vec{w} + \lambda \vec{w}^T \vec{w}$$

$$\Rightarrow \nabla_w L_{\text{ridge}}(w) = 2 X X^T w + 2 \lambda w - 2 X \vec{y}$$

Equating to zero

$$X X^T w + \lambda I w = X \vec{y}$$

$$\Rightarrow (X X^T + \lambda I) w = X \vec{y}$$

$$\Rightarrow \hat{w} = (X X^T + \lambda I)^{-1} X \vec{y}$$

For $\lambda > 0$
 $(X X^T + \lambda I)$
is ALWAYS
invertible

Probabilistic (Bayesian) view of regularized linear regression

$$L_{\text{ridge}}(w) = \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2 + \lambda \|w\|_2^2$$

Bringing back the -ve sign as in case of the ^{log} likelihood fn

$$\log(L_{\text{ridge}}(w)) = - \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2 - \lambda w^T w$$

Exponentiating RHS above

$$L_{\text{ridge}}(w) = \exp\left(- \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2\right) \cdot \exp(-\lambda w^T w)$$
$$= \left[\prod_{i=1}^N e^{-(y^{(i)} - w^T x^{(i)})^2} \right] \cdot \underbrace{e^{-\lambda w^T w}}_{\text{prior}}$$

likelihood term from
MLE (excluding the constant terms)
 σ^2 & normalizⁿ.

Remember Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$A \rightarrow$ model parameters (w), $B \rightarrow$ Data (S)

$$P(w|S) = \frac{P(S|w)P(w)}{\boxed{P(S)}} \leftarrow \begin{array}{l} \text{independent of} \\ \text{the model parameters} \\ w \therefore \text{constant} \end{array}$$

$S = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$

$$\Rightarrow \underbrace{P(w|S)}_{\text{posterior}} \propto \underbrace{P(S|w)}_{\text{likelihood}} \cdot \underbrace{P(w)}_{\text{prior}}$$

Since we assumed i.i.d samples, our likelihood had the form

$$P(S|w) = P(\epsilon^{(1)}, \epsilon^{(2)}, \dots, \epsilon^{(N)} | w) \propto \prod_{i=1}^N \exp\left(-\frac{(y^{(i)} - w^T x^{(i)})^2}{2\sigma^2}\right)$$

This likelihood quantifies how well do the model parameters fit the data. It is essentially a data fidelity term.

$$P(w) \propto e^{-\lambda w^T w} \Rightarrow w \sim \mathcal{N}(0, \lambda I)$$

The model parameters w follow a prior distribution that is zero mean, Gaussian with a covariance of λI

Different Regularizers imply different priors

$$L_{\text{lasso}}(w) = \sum_{i=1}^N (y^{(i)} - w^T x^{(i)})^2 + \lambda \|w\|_1$$
$$\Rightarrow P(w) \propto e^{-\lambda \|w\|_1}$$

$\Rightarrow P(w)$ follows a Laplacian prior

Ridge Regression $\rightarrow$ Gaussian Prior $\rightarrow \|w\|_2^2$ regularizer

Lasso Regression $\rightarrow$ Laplacian Prior $\rightarrow \|w\|_1$ regularizer

Elastic-Net $\rightarrow$ Gaussian & Laplacian mixture Prior

$$\rightarrow \lambda_1 \|w\|_1 + \lambda_2 \|w\|_2^2$$

two regularizer terms.

The likelihood term remains the same
— essentially the data fidelity term.

For regularized models, we use the **Maximum a Posteriori (MAP)** estimate as opposed to the **Maximum Likelihood Estimate (MLE)**, which only worries about fitting the data well.

As we have seen in the class, regularization helps in reducing variance in the model parameter estimation and therefore helps avoid overfitting.
The price we pay is an increase in Bias